Developing the Olist dashboard in Power BI: An analytical journey

Olist S.A. is a Brazilian e-commerce marketplace that connects small businesses with sales channels across the country. This project aimed to improve logistics and commercial decision-making in order to enhance the end-customer experience through data analysis.

by María del Mar Oli

Data collection and preparation

- **Source identification:** Extraction of the public **Olist dataset** from Kaggle, covering customer, seller, product, geolocation, and review data.
- **Data cleansing:** Removal of duplicates and handling of null or outlier values in delivery dates and review scores.
- **Transformation:** Use of Power Query to standardize date formats and create critical calculated fields, such as delivery times in days.
- **Model loading:** Import of a clean and unified dataset into Power BI.

	Vacío 0%	Vacío 0%	Vacío 0%	Vacío 0
	Distintos: 1000; únicos: 1000	Distintos: 1000; únicos: 1000	Distintos: 5; únicos: 0	Distintos: 1000; únicos: 1000
1	00010242fe8c5a6d1ba2dd792cb162...	3ce436f183e68e07877b285a838db1...	delivered	13/09/2017 08:59:
2	73fc7af87114b39712e6da79b0a377...	41dcb106f807e993532d4462632901...	delivered	11/01/2018 15:30:
3	b81ef226f3fe1789b1e8b2acac839d17	0a8556ac6be836b46b3e89920d592...	delivered	25/04/2018 22:01:
4	a548910a1c6147796b98fdf73dbeba...	8a2e7ef9053dea531e4dc76bd6d853...	delivered	28/02/2018 12:25:
5	53cdb2fc8bc7dce0b6741e21502734...	b0830fb4747a6c6d20dea0b8c802d7...	delivered	24/07/2018 20:41:
6	a9810da82917af2d9aefd1278f1dcfa0	f2c7fc58a9de810828715166c672f10a	delivered	26/06/2018 11:01:
7	658677c97b385a9be170737859d35...	de6dff97e5f1ba84a3cd9a3bc97df5f6	delivered	09/04/2017 17:41:
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9	25e8ea4e93396b6fa0d3dd708e76c1...	25b14b69de0b6e184ae6fe2755e47...	delivered	12/12/2017 11:19:
10	b18dcdf73be66366873cd26c5724d1...	eecafc3ff695f031bfe354a9fff9d437	delivered	06/04/2018 22:18:
11	47770eb9100c2d0c44946d9cf07ec6...	41ce2a54c0b03bf3443c3d931a3670...	delivered	08/08/2018 08:38:
12	e48aa0d2dcec3a2e87348811bcfdf22b	840be0730010e9580efd289bc9f377...	delivered	30/06/2017 15:38:
13	ba78997921bbcdc1373bb41e913ab...	7a5d8efaaa1081f800628c30d2b0728f	delivered	06/12/2017 12:04:
14	c31a859e34e3adac22f376954e19b3...	98f485515e575607a8003b17e91990...	delivered	07/08/2018 23:12:
15	000229ec398224ef6ca0657da4fc703e	6489ae5e4333f3693df5ad4372dab6...	delivered	14/01/2018 14:33:
16	9c214ac970e84273583ab523dfafd09b	daf813bc2e41c86027d086e8dfb355...	delivered	08/05/2017 13:35:
17	42fdf880ba16b47b59251dd489d444...	15fd6fb8f8312dbb4674e4518d6fa3b3	delivered	21/05/2018 13:59:
18	b9bf720beb4ab3728760088589c621...	a5224bdc7685fd39cd7a2340441549...	delivered	14/05/2018 10:29:
19	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d...	delivered	18/11/2017 19:28:
20	298fcdf1f73eb413e4d26d01b25bc1cd	a24e6f72471e9dbafcb292bc318f4859	delivered	07/05/2018 13:20:
21	3d374c9e46530bb5ed4a764891530...	b1af08103e1e17838522e5b9fbe2ed...	delivered	06/12/2017 10:36:

Data modeling and DAX

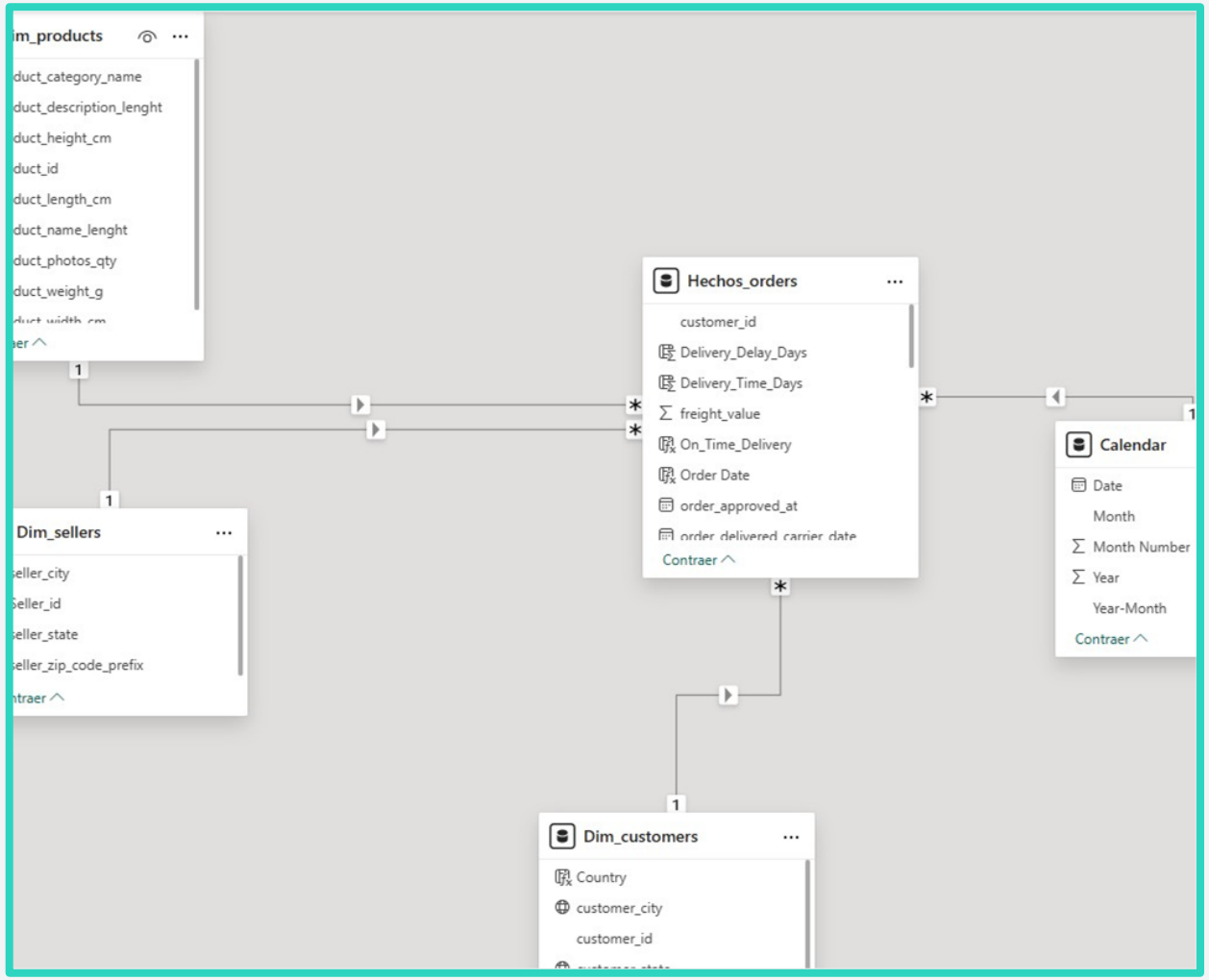

Table relationships: Establishment of a star schema connecting the fact table (orders) with dimension tables (customers, products, and others).

DAX calculations: Creation of advanced custom measures such as delayed order percentage, average times, and positive review percentage.

Hierarchies: Implementation of date hierarchies (Year-Month) and geographic hierarchies (Country > State > City) to enable multi-level analysis.

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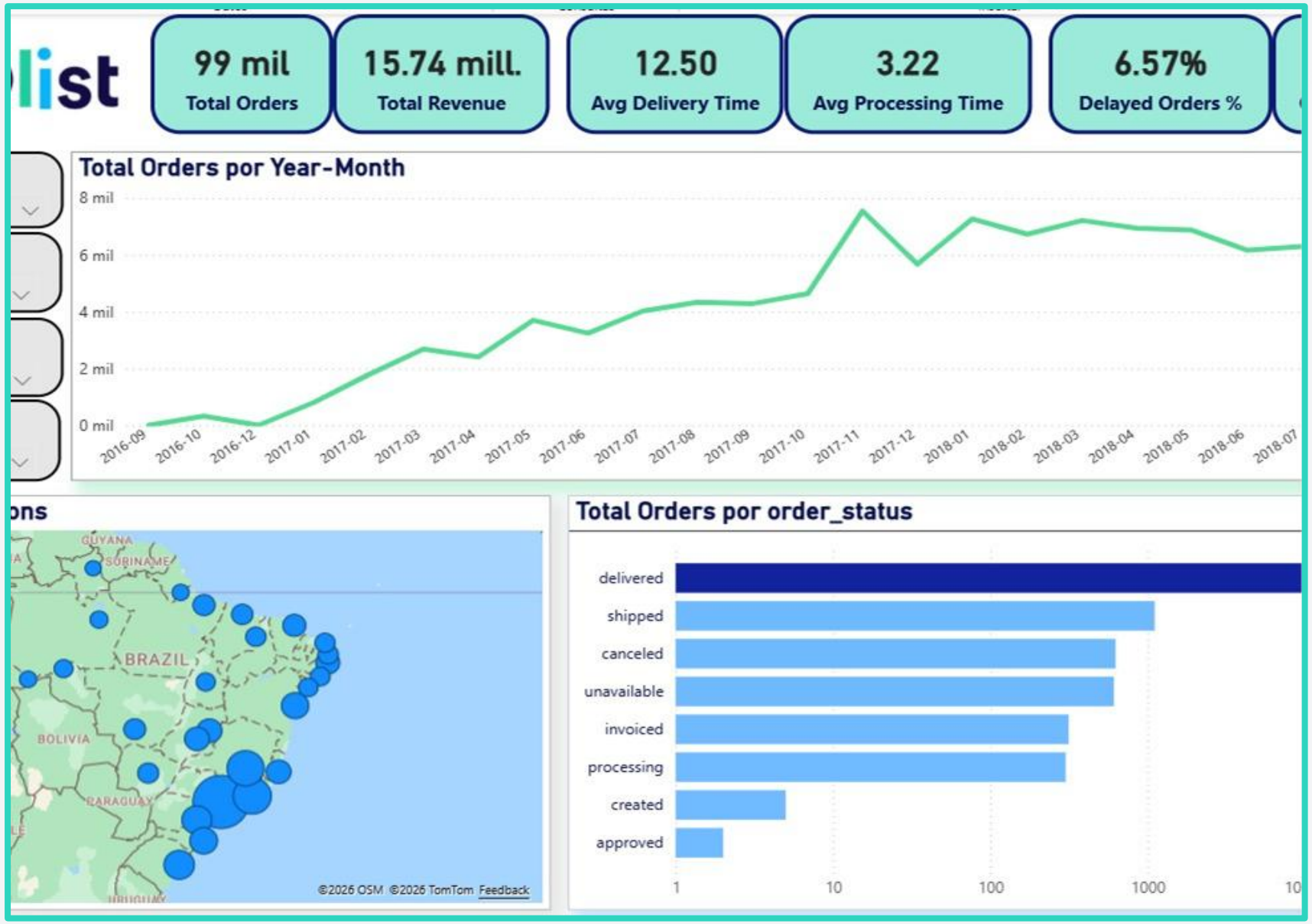
Logistics overview

The first level of the dashboard provides a high-level view of the business. It tracks key KPIs such as total order volume, average delivery time, and on-time delivery rates.

The time trend helps identify demand peaks, while the geographic map clearly shows that most operations are concentrated in Brazil's coastal states.



Operational analysis and bottlenecks

This section explores the root causes of logistics issues

Using scatter plots and conditionally formatted matrices with data bars, we can quickly identify which product categories, sellers, and geographic regions are experiencing the longest delays.

This helps logistics management focus improvement efforts where they will have the greatest impact on the supply chain.

Customer impact

The final level of the dashboard connects operational performance with the end customer. It analyzes the distribution of review scores from 1 to 5 stars and how those ratings evolve over time.

The **key finding** is clear: Customer satisfaction drops sharply when an order is delayed.

The choropleth map makes it easy to visualize customer satisfaction across different regions of Brazil.

Interface design and usability

Color palette

Selection of corporate blue and green tones, supported by semantic colors such as green and red for performance alerts.

Spatial distribution

KPIs are placed in the upper area of the dashboard, following an F-pattern reading flow and reinforcing a visual hierarchy from general to specific.

Typography

Clean fonts and the removal of visual noise were used to enable fast and effective reading.

Interactivity implementation

Dynamic Filters and Slicers

Interactivity turns a static dashboard into a powerful analytical tool. Dynamic slicers such as Date, Order Status, and Customer State were implemented in the side panel and affect the entire model.

In addition, all visual elements are interconnected: when a user selects a state on the map or a problematic product category in a chart, the rest of the dashboard automatically recalculates to display only the relevant information for that selection.

Tooltips: Depth without clutter

Tooltips allow users to access detailed and contextual information by hovering over an element, without the need to change pages.

> *"When hovering over a category or a state, a pop-up window displays a visual summary of sales, average rating, and delay breakdown"*

Conclussion and next steps

- The dashboard transforms thousands of rows of data into effective real-time logistics decisions.
- Management now has an intuitive tool that connects operational performance with final customer satisfaction.
- **Next step:** Implement predictive analytics using Machine Learning to anticipate stockouts and seasonal delays.

With this analytical model, Olist S.A. now has a connected and end-to-end view of its value chain, capable of identifying operational inefficiencies before they critically affect customer satisfaction.